Sample Composition and Habitat Distribution

Species \ Island	Biscoe	Dream	Torgersen
Adelie	44	55	47
Chinstrap	0	68	0
Gentoo	119	0	0

Table 1: Species Distribution Across Islands

A total of 333 penguins (after excluding missing observations) were included in the analysis, comprising three species: Adelie (n = 146), Gentoo (n = 119), and Chinstrap (n = 68). As summarized in **Table 1**, the frequency distribution across islands revealed distinct habitat preferences: Adelie penguins were distributed across all three islands (Biscoe: n = 44, Dream: n = 55, Torgersen: n = 47), whereas Chinstrap penguins were found exclusively on Dream Island (n = 68) and Gentoo penguins exclusively on Biscoe Island (n = 119).

Morphological Differences in Bill Dimensions

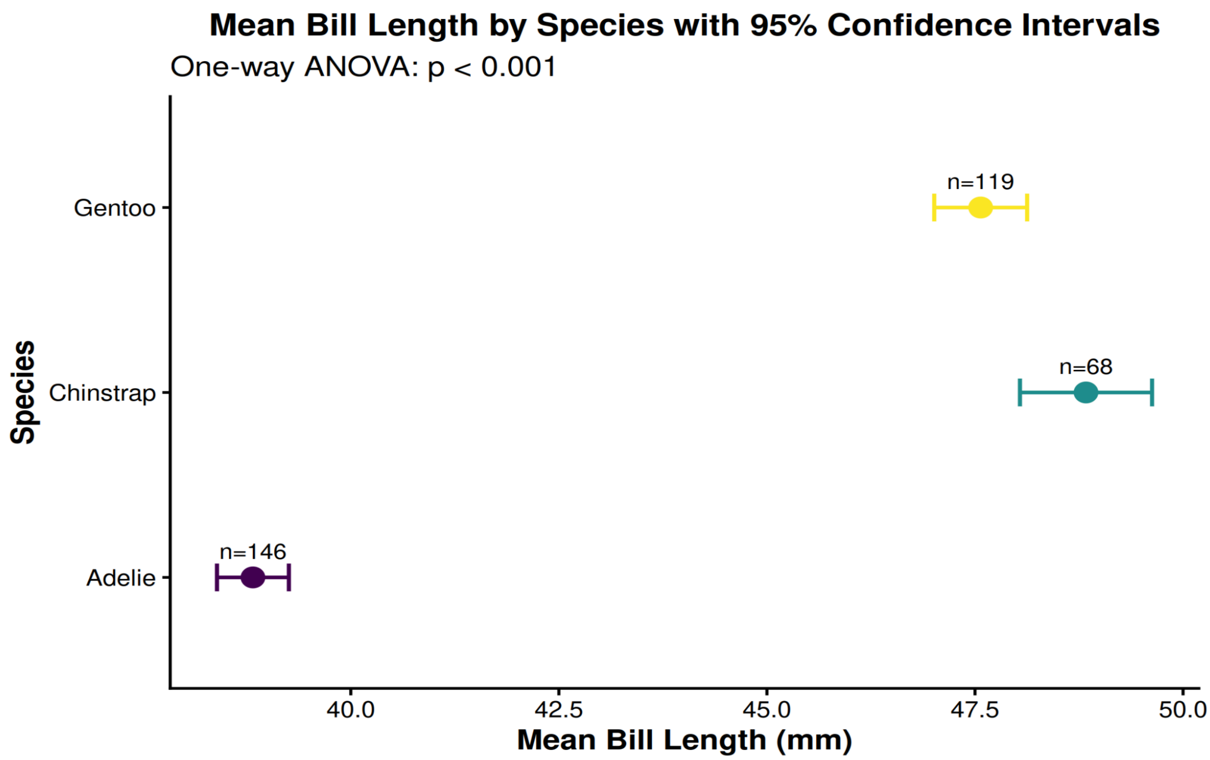

Figure 1: Mean Bill Length by Species with 95% Confidence Intervals

A one-way ANOVA revealed a statistically significant difference in mean bill length among the three species ($p < 0.001$). As illustrated in the forest plot (**Figure 1**), Chinstrap penguins had the highest mean bill length (≈48.8 mm), followed by Gentoo (≈47.5 mm), while Adelie penguins exhibited the shortest mean bill length (≈38.8 mm). The non-overlapping confidence intervals between Adelie and the other two species confirm a clear morphological separation, while the confidence intervals of Chinstrap and Gentoo showed partial overlap, suggesting a comparatively smaller difference between these two species.

Body Mass and Sexual Dimorphism

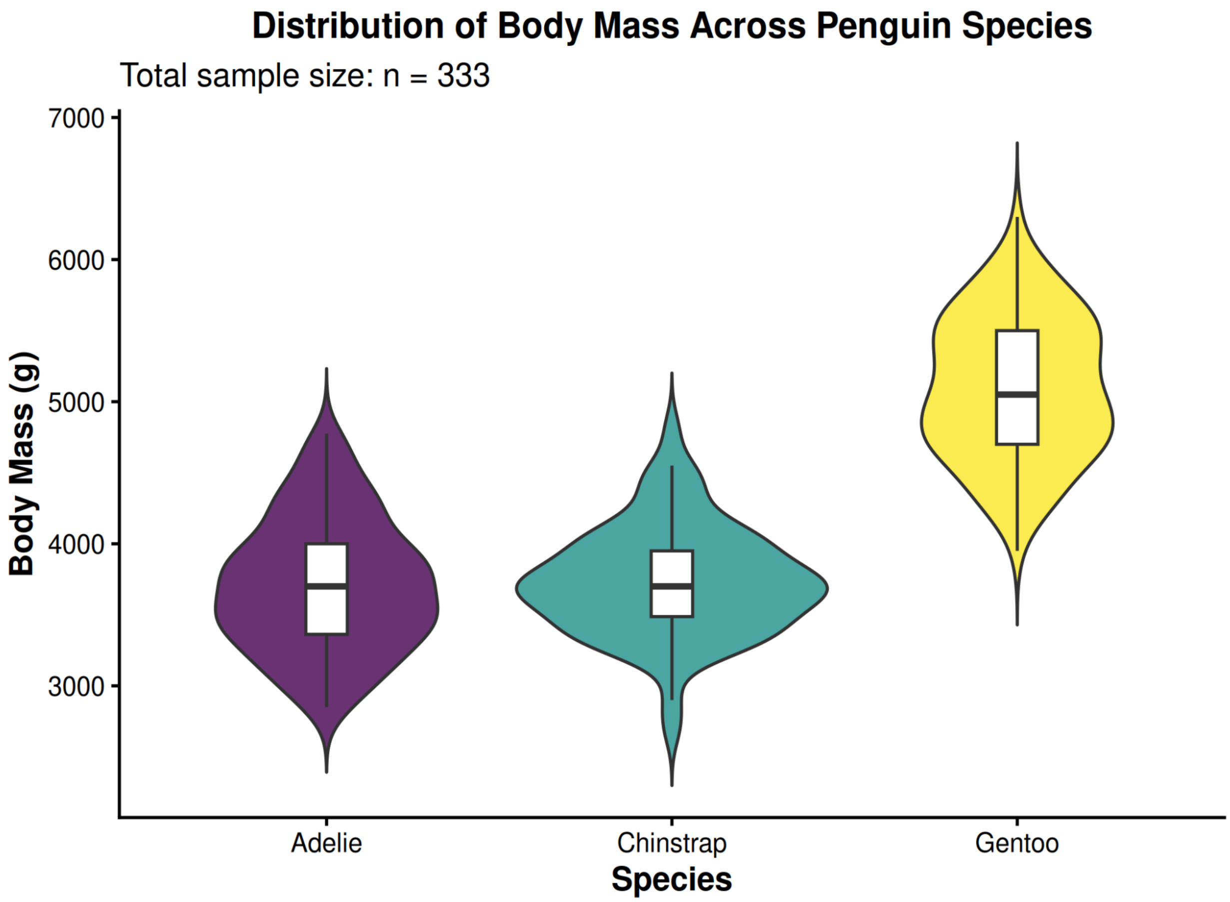

Figure 2: Distribution of Body Mass Across Penguin Species
Figure 3: Body Mass Distribution by Species and Sex

Body mass distributions, visualized through violin plots (**Figure 2**), showed that Gentoo penguins were markedly heavier (median ≈ 5,050 g) than Adelie and Chinstrap (median ≈ 3,700–3,750 g), and also displayed the widest spread, indicating greater variability in body size within this species. Furthermore, sex-disaggregated boxplots (**Figure 3**) revealed consistent sexual dimorphism across all species, with male penguins showing higher median body mass than females in every group. This difference was most pronounced in Gentoo, where males (median ≈ 5,500 g) were substantially heavier than females (median ≈ 4,700 g), while Adelie and Chinstrap showed a comparatively smaller, though still consistent, male-female gap.

Bivariate Analysis of Bill Morphology

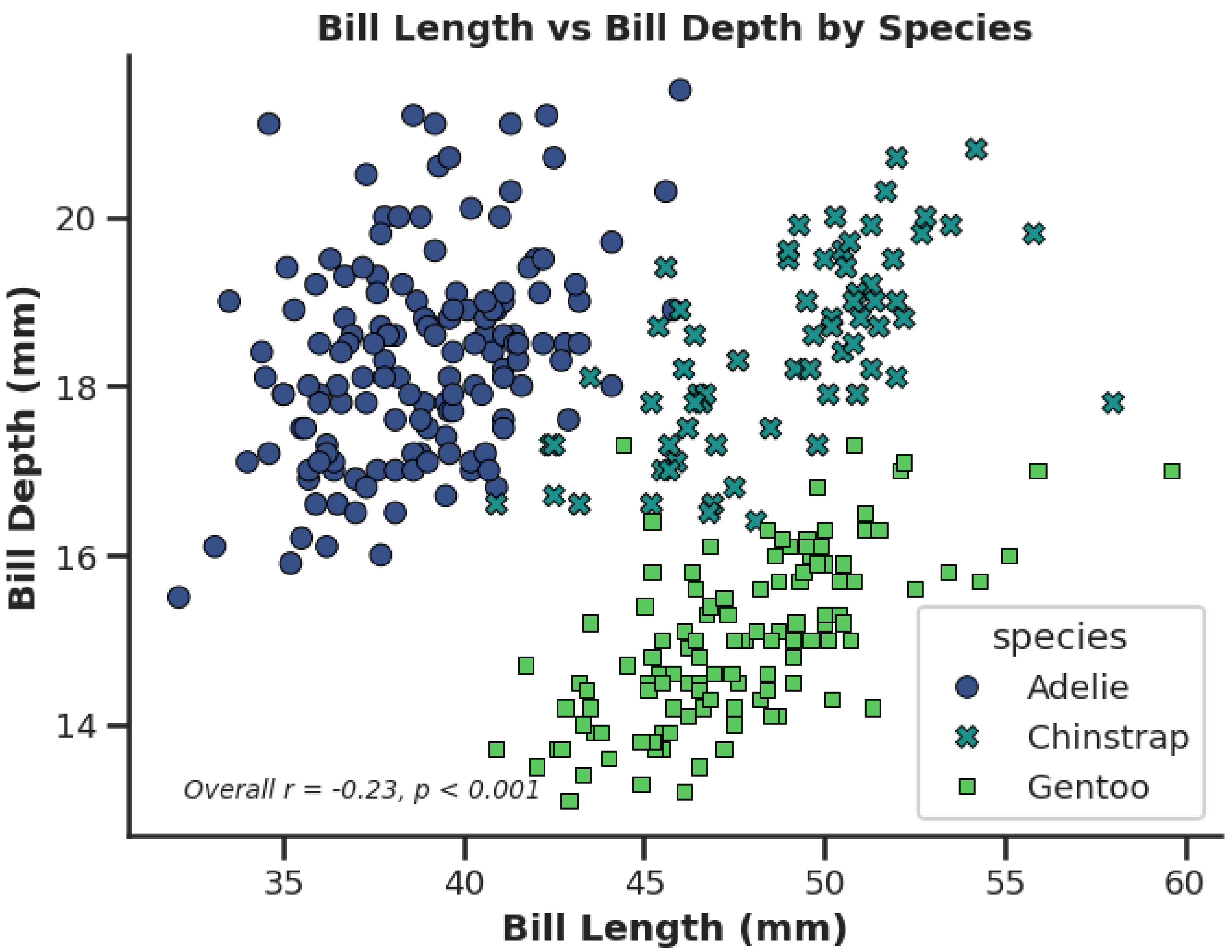

Figure 4: Bill Length vs Bill Depth by Species

Bivariate analysis of bill length and bill depth revealed a weak overall correlation between the two measurements ($r = -0.229$, $p < 0.001$). Despite this weak pooled correlation, the scatter plot (**Figure 4**) showed clear species-level clustering, with Adelie occupying a region of shorter bill length but relatively higher bill depth, Gentoo characterized by longer bills but lower bill depth, and Chinstrap positioned at intermediate-to-high values for both measurements. This clustering pattern suggests that bill morphology, when considered alongside species identity, provides meaningful discrimination among the three penguin groups, even though the pooled linear correlation alone is weak.

Summary

Overall, sample composition was uneven across species (Adelie: 146; Gentoo: 119; Chinstrap: 68), which should be considered when interpreting the relative precision of group-wise estimates. Collectively, these findings demonstrate significant interspecies morphological differentiation in both bill dimensions and body mass, along with pronounced sexual dimorphism, consistent with known ecological and species-specific adaptations among Pygoscelis penguins in the Palmer Archipelago.